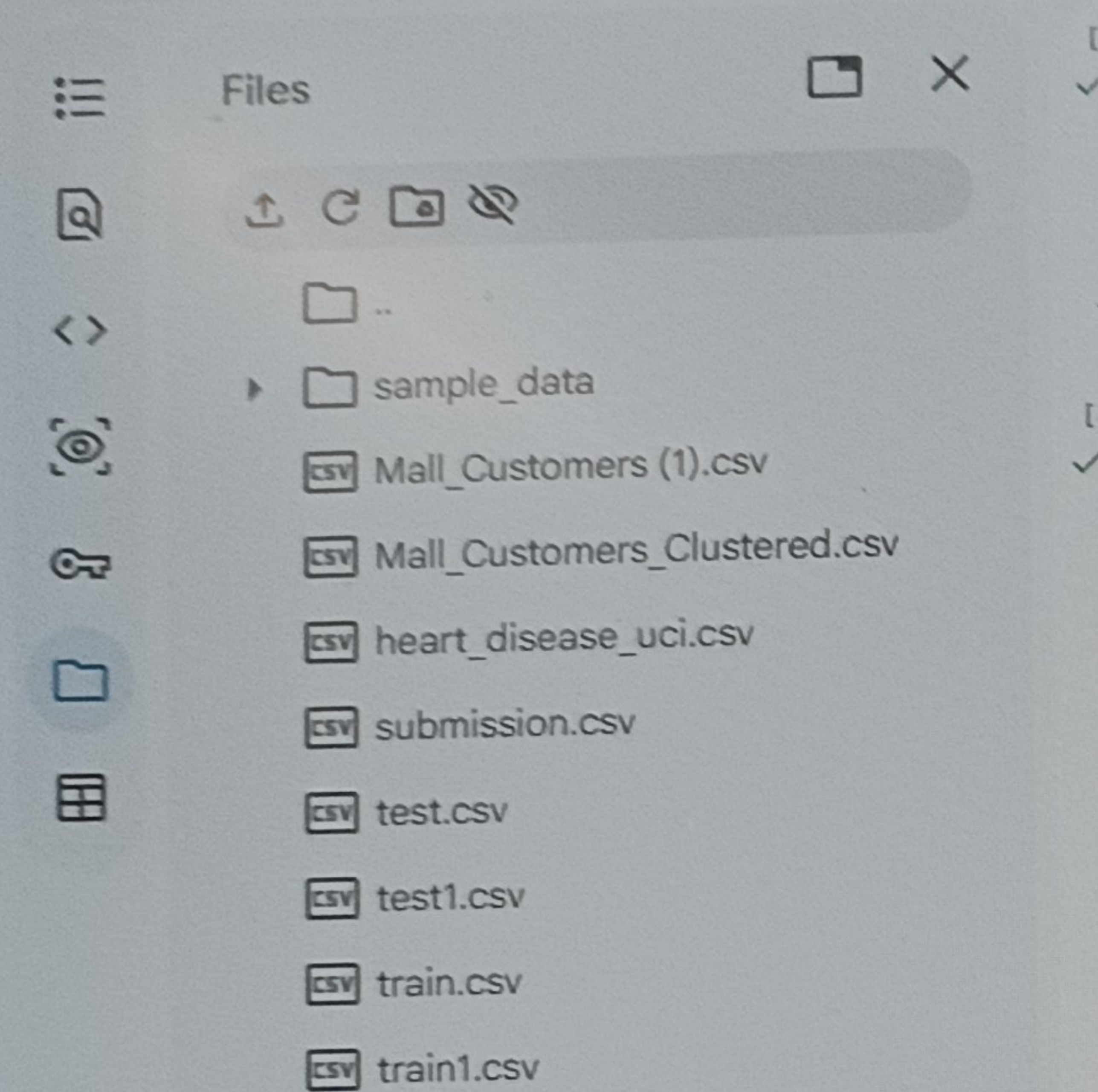




```
output.to_csv('submission.csv', index=False)

print("Submission file created successfully!")
```

Submission file created successfully!

```
[10]
# Import libraries
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans

# Upload dataset
from google.colab import files
uploaded = files.upload()

# Read dataset
df = pd.read_csv('Mall_Customers.csv')

# Display first 5 rows
print(df.head())

# Select features for clustering
X = df[['Annual Income (k$)', 'Spending Score (1-100)']]

# Find optimal clusters using Elbow Method
wcss = []

for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=42, n_init=10)
    kmeans.fit(X)
    wcss.append(kmeans.inertia_)

plt.figure(figsize=(8,5))
plt.plot(range(1,11), wcss, marker='o')
plt.title('Elbow Method')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')
plt.show()
```